

Source folder: /Users/jonat/Documents/GitHub/master_thesis
Files in this part: 17

File: federated_learning_in_mobile/test_driver/integration_test.dart

```
import 'dart:io';

import 'package:integration_test/integration_test_driver_extended.dart';

const String _outputDirectoryPath = 'artifacts/integration_screenshots';

Future<void> main() async {
  final outputDirectory = Directory(_outputDirectoryPath);
  await outputDirectory.create(recursive: true);

  await integrationDriver(onScreenshot: _saveScreenshot);
}

Future<bool> _saveScreenshot(
  String screenshotName,
  List<int> screenshotBytes, [
  Map<String, Object?>? args,
]) async {
  final screenshotFile = File('$_outputDirectoryPath/$screenshotName.png');
  await screenshotFile.parent.create(recursive: true);
  await screenshotFile.writeAsBytes(screenshotBytes);
  stdout.writeln('Saved screenshot to ${screenshotFile.path}');
  return true;
}
```

File: federated_learning_in_mobile/test_driver/screenshot_test_driver.dart

```
import 'dart:io';

import 'package:integration_test/integration_test_driver_extended.dart';

const String _outputDirectoryPath = 'artifacts/integration_screenshots';

Future<void> main() async {
  final outputDirectory = Directory(_outputDirectoryPath);
  await outputDirectory.create(recursive: true);

  await integrationDriver(onScreenshot: _saveScreenshot);
}

Future<bool> _saveScreenshot(
  String screenshotName,
  List<int> screenshotBytes, [
  Map<String, Object?>? args,
]) async {
  final screenshotFile = File('$_outputDirectoryPath/$screenshotName.png');
  await screenshotFile.parent.create(recursive: true);
  await screenshotFile.writeAsBytes(screenshotBytes);
  stdout.writeln('Saved screenshot to ${screenshotFile.path}');
  return true;
}
```

File: federated_learning_in_mobile/tooling/build_movie_titles_asset.py

```
import csv
from pathlib import Path

ROOT = Path(__file__).resolve().parents[2]
source = ROOT / "ml-100k" / "u.item"
out_dir = ROOT / "federated_learning_in_mobile" / "assets"
out_dir.mkdir(parents=True, exist_ok=True)
out_file = out_dir / "movielens_titles.csv"
```

```

with source.open("r", encoding="latin-1") as src, out_file.open("w", encoding="utf-8", newline="") as dst:
    writer = csv.writer(dst)
    writer.writerow(["item_id", "title"])
    count = 0
    for line in src:
        parts = line.rstrip("\n").split("|")
        if len(parts) < 2 or not parts[0].isdigit():
            continue
        item_id = int(parts[0]) - 1
        writer.writerow([item_id, parts[1]])
        count += 1

print(f"Wrote {count} movie titles to {out_file}")

```

File: figures/summary_table.csv

```

Experiment,Epsilon,Alpha,NDCG@10,Hit@10,Accuracy
Centralized Baseline,N/A,N/A,0.22501065427109246,0.38,0.05495
DP Sweep,1,0.5,0.0467 ± 0.0121,0.0533 ± 0.0047,0.0000 ± 0.0000
DP Sweep,2,0.5,0.0456 ± 0.0095,0.0567 ± 0.0094,0.0000 ± 0.0000
DP Sweep,4,0.5,0.0479 ± 0.0091,0.0600 ± 0.0082,0.0000 ± 0.0000
DP Sweep,8,0.5,0.0534 ± 0.0131,0.0600 ± 0.0082,0.0000 ± 0.0000
DP Sweep,∞,0.5,0.0539 ± 0.0108,0.0633 ± 0.0205,0.0000 ± 0.0000
Heterogeneity Sweep,∞,0.1,0.0538 ± 0.0108,0.0633 ± 0.0205,0.0000 ± 0.0000
Heterogeneity Sweep,∞,0.5,0.0539 ± 0.0108,0.0633 ± 0.0205,0.0000 ± 0.0000
Heterogeneity Sweep,∞,1.0,0.0539 ± 0.0108,0.0633 ± 0.0205,0.0000 ± 0.0000

```

File: mobile_results/dp_inf_dim_16_clients_1_device_Android_Android_SDK_built_for_x86_64_t1773173944932.json

```

{
  "experiment_id": "dp_inf_dim_16_clients_1_device_Android_Android_SDK_built_for_x86_64_t1773173944932",
  "timestamp": "2026-03-11T01:49:04.932940",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 16,
    "client_id": "android_client_6213",
    "device_id": "Android_Android_SDK_built_for_x86_64",
    "server_url": "http://10.0.2.2:8080"
  },
  "rounds": [
    {
      "round": 0,
      "train_loss": 1.0370102800724106,
      "test_metrics": {
        "accuracy": 0.0,
        "mse": 1.0363101971409518,
        "mae": 1.0174406413106594,
        "Hit@10": 0.0,
        "NDCG@10": 0.0,
        "Precision@10": 0.0,
        "Recall@10": 0.0,
        "samples": 10
      },
      "aggregation": {
        "num_clients": 1,
        "total_samples": 10
      },
      "client_metrics": [
        {
          "client_id": "android_client_6213",
          "loss": 1.0370102800724106,
          "samples": 10,
          "epochs": 1
        }
      ],
      "resource_metrics": {
        "battery_level": 100,
        "battery_drain": 0.0,
        "elapsed_seconds": 5,
        "device_info": {
          "platform": "Android",
          "model": "Android SDK built for x86_64",

```

```
    "manufacturer": "unknown",
    "version": "16",
    "sdkInt": 36
  },
  "timestamp": "2026-03-11T01:49:10.268030",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2026-03-11T01:49:10.429509"
},
{
  "round": 1,
  "train_loss": 1.0370102800724106,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 1.0363101971409516,
    "mae": 1.0174406413106591,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_6213",
      "loss": 1.0370102800724106,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 6,
    "device_info": {
      "platform": "Android",
      "model": "Android SDK built for x86_64",
      "manufacturer": "unknown",
      "version": "16",
      "sdkInt": 36
    },
    "timestamp": "2026-03-11T01:49:11.468666",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2026-03-11T01:49:11.663107"
},
{
  "round": 2,
  "train_loss": 1.0370102800724106,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 1.0363101971409516,
    "mae": 1.0174406413106594,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_6213",
      "loss": 1.0370102800724106,
      "samples": 10,
```

```
    "epochs": 1
  },
  {
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 7,
      "device_info": {
        "platform": "Android",
        "model": "Android SDK built for x86_64",
        "manufacturer": "unknown",
        "version": "16",
        "sdkInt": 36
      },
      "timestamp": "2026-03-11T01:49:12.563101",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2026-03-11T01:49:12.704573"
  },
  {
    "round": 3,
    "train_loss": 1.0370102800724106,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 1.0363101971409518,
      "mae": 1.0174406413106594,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    },
    "aggregation": {
      "num_clients": 1,
      "total_samples": 10
    },
    "client_metrics": [
      {
        "client_id": "android_client_6213",
        "loss": 1.0370102800724106,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 8,
      "device_info": {
        "platform": "Android",
        "model": "Android SDK built for x86_64",
        "manufacturer": "unknown",
        "version": "16",
        "sdkInt": 36
      },
      "timestamp": "2026-03-11T01:49:13.572915",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2026-03-11T01:49:13.710113"
  },
  {
    "round": 4,
    "train_loss": 1.0370102800724106,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 1.0363101971409516,
      "mae": 1.0174406413106594,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    }
```

```
    },
    "aggregation": {
      "num_clients": 1,
      "total_samples": 10
    },
    "client_metrics": [
      {
        "client_id": "android_client_6213",
        "loss": 1.0370102800724106,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 9,
      "device_info": {
        "platform": "Android",
        "model": "Android SDK built for x86_64",
        "manufacturer": "unknown",
        "version": "16",
        "sdkInt": 36
      },
      "timestamp": "2026-03-11T01:49:14.574457",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2026-03-11T01:49:14.720162"
  },
  {
    "round": 5,
    "train_loss": 1.0370102800724106,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 1.0363101971409518,
      "mae": 1.0174406413106591,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    },
    "aggregation": {
      "num_clients": 1,
      "total_samples": 10
    },
    "client_metrics": [
      {
        "client_id": "android_client_6213",
        "loss": 1.0370102800724106,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 10,
      "device_info": {
        "platform": "Android",
        "model": "Android SDK built for x86_64",
        "manufacturer": "unknown",
        "version": "16",
        "sdkInt": 36
      },
      "timestamp": "2026-03-11T01:49:15.626569",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2026-03-11T01:49:15.750405"
  },
  {
    "round": 6,
```

```
"train_loss": 1.0370102800724106,
"test_metrics": {
  "accuracy": 0.0,
  "mse": 1.0363101971409518,
  "mae": 1.0174406413106591,
  "Hit@10": 0.0,
  "NDCG@10": 0.0,
  "Precision@10": 0.0,
  "Recall@10": 0.0,
  "samples": 10
},
"aggregation": {
  "num_clients": 1,
  "total_samples": 10
},
"client_metrics": [
  {
    "client_id": "android_client_6213",
    "loss": 1.0370102800724106,
    "samples": 10,
    "epochs": 1
  }
],
"resource_metrics": {
  "battery_level": 100,
  "battery_drain": 0.0,
  "elapsed_seconds": 11,
  "device_info": {
    "platform": "Android",
    "model": "Android SDK built for x86_64",
    "manufacturer": "unknown",
    "version": "16",
    "sdkInt": 36
  },
  "timestamp": "2026-03-11T01:49:16.610158",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2026-03-11T01:49:16.726557"
},
{
  "round": 7,
  "train_loss": 1.0370102800724106,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 1.0363101971409518,
    "mae": 1.0174406413106591,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_6213",
      "loss": 1.0370102800724106,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 12,
    "device_info": {
      "platform": "Android",
      "model": "Android SDK built for x86_64",
      "manufacturer": "unknown",
      "version": "16",
```

```
    "sdkInt": 36
  },
  "timestamp": "2026-03-11T01:49:17.628607",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2026-03-11T01:49:17.805540"
},
{
  "round": 8,
  "train_loss": 1.0370102800724106,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 1.0363101971409516,
    "mae": 1.0174406413106591,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_6213",
      "loss": 1.0370102800724106,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 13,
    "device_info": {
      "platform": "Android",
      "model": "Android SDK built for x86_64",
      "manufacturer": "unknown",
      "version": "16",
      "sdkInt": 36
    },
    "timestamp": "2026-03-11T01:49:18.683363",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2026-03-11T01:49:18.857391"
},
{
  "round": 9,
  "train_loss": 1.0370102800724106,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 1.0363101971409518,
    "mae": 1.017440641310659,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_6213",
      "loss": 1.0370102800724106,
      "samples": 10,
      "epochs": 1
    }
  ]
}
```

```
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 14,
      "device_info": {
        "platform": "Android",
        "model": "Android SDK built for x86_64",
        "manufacturer": "unknown",
        "version": "16",
        "sdkInt": 36
      },
      "timestamp": "2026-03-11T01:49:19.688237",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2026-03-11T01:49:19.891978"
  }
],
"final_metrics": {},
"client_metrics": [],
"mobile_metrics": [],
"server_received_at": "2026-03-11T01:49:22.465230"
}
```

File: mobile_results/dp_inf_dim_16_clients_1_device_Android_Android_SDK_built_for_x86_64_t1773173944932_sum

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients>Total_Samples,Training_
Time_MS,Battery_Drain
0,1.0370102800724106,0.0,0.0,0.0,0.0,1.0363101971409518,1.0174406413106594,0.0,1,10,,0.0
1,1.0370102800724106,0.0,0.0,0.0,0.0,1.0363101971409516,1.0174406413106591,0.0,1,10,,0.0
2,1.0370102800724106,0.0,0.0,0.0,0.0,1.0363101971409516,1.0174406413106594,0.0,1,10,,0.0
3,1.0370102800724106,0.0,0.0,0.0,0.0,1.0363101971409518,1.0174406413106594,0.0,1,10,,0.0
4,1.0370102800724106,0.0,0.0,0.0,0.0,1.0363101971409516,1.0174406413106594,0.0,1,10,,0.0
5,1.0370102800724106,0.0,0.0,0.0,0.0,1.0363101971409518,1.0174406413106591,0.0,1,10,,0.0
6,1.0370102800724106,0.0,0.0,0.0,0.0,1.0363101971409518,1.0174406413106591,0.0,1,10,,0.0
7,1.0370102800724106,0.0,0.0,0.0,0.0,1.0363101971409518,1.0174406413106591,0.0,1,10,,0.0
8,1.0370102800724106,0.0,0.0,0.0,0.0,1.0363101971409516,1.0174406413106591,0.0,1,10,,0.0
9,1.0370102800724106,0.0,0.0,0.0,0.0,1.0363101971409518,1.017440641310659,0.0,1,10,,0.0
FINAL,,,,,,,,,
```

File: mobile_results/dp_inf_dim_16_clients_1_device_google_sdk_gphone64_arm64_t1773794443844.json

```
{
  "experiment_id": "dp_inf_dim_16_clients_1_device_google_sdk_gphone64_arm64_t1773794443844",
  "timestamp": "2026-03-18T06:10:43.844865",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 16,
    "client_id": "android_client_4439",
    "device_id": "google_sdk_gphone64_arm64",
    "server_url": "http://10.0.2.2:8000"
  },
  "rounds": [
    {
      "round": 0,
      "train_loss": 0.9859451509076628,
      "test_metrics": {
        "accuracy": 0.0,
        "mse": 0.9855484210734814,
        "mae": 0.992234706268677,
        "Hit@10": 0.0,
        "NDCG@10": 0.0,
        "Precision@10": 0.0,
        "Recall@10": 0.0,
        "samples": 15
      },
      "aggregation": {
        "num_clients": 1,
        "total_samples": 15
      },
      "client_metrics": [
        {
```



```
    "client_id": "android_client_4439",
    "loss": 0.9859451509076628,
    "samples": 15,
    "epochs": 1
  },
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 4,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_arm64",
      "manufacturer": "Google",
      "version": "16",
      "sdkInt": 36
    },
    "timestamp": "2026-03-18T06:10:48.115539",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2026-03-18T06:10:48.190310"
},
{
  "round": 1,
  "train_loss": 0.9859451509076628,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9855484210734814,
    "mae": 0.9922347062686769,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 15
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 15
  },
  "client_metrics": [
    {
      "client_id": "android_client_4439",
      "loss": 0.9859451509076628,
      "samples": 15,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 28,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_arm64",
      "manufacturer": "Google",
      "version": "16",
      "sdkInt": 36
    },
    "timestamp": "2026-03-18T06:11:11.876517",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2026-03-18T06:11:11.927268"
},
{
  "round": 0,
  "train_loss": 0.9859451509076629,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9855484210734816,
    "mae": 0.9922347062686772,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
```

```
"Precision@10": 0.0,
"Recall@10": 0.0,
"samples": 15
},
"aggregation": {
  "num_clients": 1,
  "total_samples": 15
},
"client_metrics": [
  {
    "client_id": "android_client_4439",
    "loss": 0.9859451509076629,
    "samples": 15,
    "epochs": 1
  }
],
"resource_metrics": {
  "battery_level": 100,
  "battery_drain": 0.0,
  "elapsed_seconds": 31,
  "device_info": {
    "platform": "Android",
    "model": "sdk_gphone64_arm64",
    "manufacturer": "Google",
    "version": "16",
    "sdkInt": 36
  },
  "timestamp": "2026-03-18T06:11:15.433001",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2026-03-18T06:11:15.483744"
},
{
  "round": 1,
  "train_loss": 0.985945150907663,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9855484210734815,
    "mae": 0.9922347062686772,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 15
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 15
  },
  "client_metrics": [
    {
      "client_id": "android_client_4439",
      "loss": 0.985945150907663,
      "samples": 15,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 32,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_arm64",
      "manufacturer": "Google",
      "version": "16",
      "sdkInt": 36
    },
    "timestamp": "2026-03-18T06:11:16.165882",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2026-03-18T06:11:16.222021"
```

```
},
{
  "round": 2,
  "train_loss": 0.9859451509076629,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9855484210734813,
    "mae": 0.992234706268677,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 15
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 15
  },
  "client_metrics": [
    {
      "client_id": "android_client_4439",
      "loss": 0.9859451509076629,
      "samples": 15,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 33,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_arm64",
      "manufacturer": "Google",
      "version": "16",
      "sdkInt": 36
    },
    "timestamp": "2026-03-18T06:11:16.861710",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2026-03-18T06:11:16.901658"
},
{
  "round": 3,
  "train_loss": 0.985945150907663,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9855484210734815,
    "mae": 0.992234706268677,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 15
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 15
  },
  "client_metrics": [
    {
      "client_id": "android_client_4439",
      "loss": 0.985945150907663,
      "samples": 15,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 33,
    "device_info": {
      "platform": "Android",
```

```
    "model": "sdk_gphone64_arm64",
    "manufacturer": "Google",
    "version": "16",
    "sdkInt": 36
  },
  "timestamp": "2026-03-18T06:11:17.585918",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2026-03-18T06:11:17.638686"
},
{
  "round": 4,
  "train_loss": 0.9859451509076629,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9855484210734815,
    "mae": 0.9922347062686772,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 15
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 15
  },
  "client_metrics": [
    {
      "client_id": "android_client_4439",
      "loss": 0.9859451509076629,
      "samples": 15,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 34,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_arm64",
      "manufacturer": "Google",
      "version": "16",
      "sdkInt": 36
    },
    "timestamp": "2026-03-18T06:11:18.285646",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2026-03-18T06:11:18.345843"
},
{
  "round": 5,
  "train_loss": 0.9859451509076629,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9855484210734814,
    "mae": 0.992234706268677,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 15
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 15
  },
  "client_metrics": [
    {
      "client_id": "android_client_4439",
      "loss": 0.9859451509076629,
```

```
        "samples": 15,
        "epochs": 1
    }
},
"resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 35,
    "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_arm64",
        "manufacturer": "Google",
        "version": "16",
        "sdkInt": 36
    },
    "timestamp": "2026-03-18T06:11:19.006585",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
},
"timestamp": "2026-03-18T06:11:19.052500"
},
{
    "round": 6,
    "train_loss": 0.9859451509076629,
    "test_metrics": {
        "accuracy": 0.0,
        "mse": 0.9855484210734815,
        "mae": 0.992234706268677,
        "Hit@10": 0.0,
        "NDCG@10": 0.0,
        "Precision@10": 0.0,
        "Recall@10": 0.0,
        "samples": 15
    },
    "aggregation": {
        "num_clients": 1,
        "total_samples": 15
    },
    "client_metrics": [
        {
            "client_id": "android_client_4439",
            "loss": 0.9859451509076629,
            "samples": 15,
            "epochs": 1
        }
    ],
    "resource_metrics": {
        "battery_level": 100,
        "battery_drain": 0.0,
        "elapsed_seconds": 35,
        "device_info": {
            "platform": "Android",
            "model": "sdk_gphone64_arm64",
            "manufacturer": "Google",
            "version": "16",
            "sdkInt": 36
        },
        "timestamp": "2026-03-18T06:11:19.755455",
        "cpu_percent": 15.0,
        "memory_mb": 50.0
    },
    "timestamp": "2026-03-18T06:11:19.794551"
},
{
    "round": 7,
    "train_loss": 0.9859451509076628,
    "test_metrics": {
        "accuracy": 0.0,
        "mse": 0.9855484210734813,
        "mae": 0.992234706268677,
        "Hit@10": 0.0,
        "NDCG@10": 0.0,
        "Precision@10": 0.0,
        "Recall@10": 0.0,
```

```
    "samples": 15
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 15
  },
  "client_metrics": [
    {
      "client_id": "android_client_4439",
      "loss": 0.9859451509076628,
      "samples": 15,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 36,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_arm64",
      "manufacturer": "Google",
      "version": "16",
      "sdkInt": 36
    },
    "timestamp": "2026-03-18T06:11:20.492746",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2026-03-18T06:11:20.536277"
},
{
  "round": 8,
  "train_loss": 0.9859451509076628,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9855484210734814,
    "mae": 0.992234706268677,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 15
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 15
  },
  "client_metrics": [
    {
      "client_id": "android_client_4439",
      "loss": 0.9859451509076628,
      "samples": 15,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 37,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_arm64",
      "manufacturer": "Google",
      "version": "16",
      "sdkInt": 36
    },
    "timestamp": "2026-03-18T06:11:21.202810",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2026-03-18T06:11:21.228094"
},
{

```

```
"round": 9,
"train_loss": 0.9859451509076629,
"test_metrics": {
  "accuracy": 0.0,
  "mse": 0.9855484210734816,
  "mae": 0.992234706268677,
  "Hit@10": 0.0,
  "NDCG@10": 0.0,
  "Precision@10": 0.0,
  "Recall@10": 0.0,
  "samples": 15
},
"aggregation": {
  "num_clients": 1,
  "total_samples": 15
},
"client_metrics": [
  {
    "client_id": "android_client_4439",
    "loss": 0.9859451509076629,
    "samples": 15,
    "epochs": 1
  }
],
"resource_metrics": {
  "battery_level": 100,
  "battery_drain": 0.0,
  "elapsed_seconds": 38,
  "device_info": {
    "platform": "Android",
    "model": "sdk_gphone64_arm64",
    "manufacturer": "Google",
    "version": "16",
    "sdkInt": 36
  },
  "timestamp": "2026-03-18T06:11:21.871233",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2026-03-18T06:11:21.898786"
},
{
  "round": 10,
  "train_loss": 0.985945150907663,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9855484210734815,
    "mae": 0.992234706268677,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 15
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 15
  },
  "client_metrics": [
    {
      "client_id": "android_client_4439",
      "loss": 0.985945150907663,
      "samples": 15,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 40,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_arm64",
      "manufacturer": "Google",
```

```
        "version": "16",
        "sdkInt": 36
    },
    "timestamp": "2026-03-18T06:11:24.778944",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
},
"timestamp": "2026-03-18T06:11:24.846776"
}
],
"final_metrics": {
    "accuracy": 0.0,
    "mse": 0.9855484210734815,
    "mae": 0.992234706268677,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "final_train_loss": 0.9859451509076627,
    "avg_accuracy": 0.0,
    "avg_Hit@10": 0.0,
    "avg_NDCG@10": 0.0
},
"client_metrics": [
    {
        "client_id": "android_client_4439",
        "total_rounds": 13,
        "total_samples": 15,
        "final_loss": 0.9859451509076627,
        "final_accuracy": 0.0,
        "timestamp": "2026-03-18T06:11:24.915701"
    }
],
"mobile_metrics": [
    {
        "device_id": "google_sdk_gphone64_arm64",
        "total_rounds": 13,
        "total_battery_drain": 0.0,
        "total_training_time_ms": 0,
        "avg_battery_drain_per_round": 0.0,
        "avg_training_time_ms_per_round": 0.0,
        "timestamp": "2026-03-18T06:11:24.916004"
    }
],
"server_received_at": "2026-03-18T06:11:26.356202"
}
```

File: mobile_results/dp_inf_dim_16_clients_1_device_google_sdk_gphone64_arm64_t1773794443844_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients,Total_Samples,Training_
Time_MS,Battery_Drain
0,0.9859451509076628,0.0,0.0,0.0,0.0,0.9855484210734814,0.992234706268677,0.0,1,15,,0.0
1,0.9859451509076628,0.0,0.0,0.0,0.0,0.9855484210734814,0.9922347062686769,0.0,1,15,,0.0
0,0.9859451509076629,0.0,0.0,0.0,0.0,0.9855484210734816,0.9922347062686772,0.0,1,15,,0.0
1,0.985945150907663,0.0,0.0,0.0,0.0,0.9855484210734815,0.9922347062686772,0.0,1,15,,0.0
2,0.9859451509076629,0.0,0.0,0.0,0.0,0.9855484210734813,0.992234706268677,0.0,1,15,,0.0
3,0.985945150907663,0.0,0.0,0.0,0.0,0.9855484210734815,0.992234706268677,0.0,1,15,,0.0
4,0.9859451509076629,0.0,0.0,0.0,0.0,0.9855484210734815,0.9922347062686772,0.0,1,15,,0.0
5,0.9859451509076629,0.0,0.0,0.0,0.0,0.9855484210734814,0.992234706268677,0.0,1,15,,0.0
6,0.9859451509076629,0.0,0.0,0.0,0.0,0.9855484210734815,0.992234706268677,0.0,1,15,,0.0
7,0.9859451509076628,0.0,0.0,0.0,0.0,0.9855484210734813,0.992234706268677,0.0,1,15,,0.0
8,0.9859451509076628,0.0,0.0,0.0,0.0,0.9855484210734814,0.992234706268677,0.0,1,15,,0.0
9,0.9859451509076629,0.0,0.0,0.0,0.0,0.9855484210734816,0.992234706268677,0.0,1,15,,0.0
10,0.985945150907663,0.0,0.0,0.0,0.0,0.9855484210734815,0.992234706268677,0.0,1,15,,0.0
FINAL,,0.0,0.0,0.0,0.0,0.9855484210734815,0.992234706268677,0.0,,,,,
```

File: mobile_results/dp_inf_dim_16_clients_1_device_google_sdk_gphone64_arm64_t1773795201066.json

```
{
    "experiment_id": "dp_inf_dim_16_clients_1_device_google_sdk_gphone64_arm64_t1773795201066",
    "timestamp": "2026-03-18T06:23:21.067231",
    "config": {
        "num_users": 943,
        "num_items": 1682,
```



```
"embedding_dim": 16,
"client_id": "android_client_520",
"device_id": "google_sdk_gphone64_arm64",
"server_url": "http://10.0.2.2:8000"
},
"rounds": [
  {
    "round": 0,
    "train_loss": 0.9923938412077214,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 0.9917313167056209,
      "mae": 0.9950340238561066,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    },
    "aggregation": {
      "num_clients": 1,
      "total_samples": 10
    },
    "client_metrics": [
      {
        "client_id": "android_client_520",
        "loss": 0.9923938412077214,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 10,
      "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_arm64",
        "manufacturer": "Google",
        "version": "16",
        "sdkInt": 36
      },
      "timestamp": "2026-03-18T06:23:31.815378",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2026-03-18T06:23:31.864244"
  },
  {
    "round": 1,
    "train_loss": 0.9923938412077214,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 0.9917313167056208,
      "mae": 0.9950340238561066,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    },
    "aggregation": {
      "num_clients": 1,
      "total_samples": 10
    },
    "client_metrics": [
      {
        "client_id": "android_client_520",
        "loss": 0.9923938412077214,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
```

```
"battery_level": 100,
"battery_drain": 0.0,
"elapsed_seconds": 11,
"device_info": {
  "platform": "Android",
  "model": "sdk_gphone64_arm64",
  "manufacturer": "Google",
  "version": "16",
  "sdkInt": 36
},
"timestamp": "2026-03-18T06:23:32.615838",
"cpu_percent": 15.0,
"memory_mb": 50.0
},
"timestamp": "2026-03-18T06:23:32.668455"
},
{
  "round": 2,
  "train_loss": 0.9923938412077214,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9917313167056209,
    "mae": 0.9950340238561066,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_520",
      "loss": 0.9923938412077214,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 12,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_arm64",
      "manufacturer": "Google",
      "version": "16",
      "sdkInt": 36
    },
    "timestamp": "2026-03-18T06:23:33.350380",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2026-03-18T06:23:33.401893"
},
{
  "round": 3,
  "train_loss": 0.9923938412077217,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9917313167056208,
    "mae": 0.9950340238561068,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
```

```
    },
    "client_metrics": [
      {
        "client_id": "android_client_520",
        "loss": 0.9923938412077217,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 13,
      "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_arm64",
        "manufacturer": "Google",
        "version": "16",
        "sdkInt": 36
      },
      "timestamp": "2026-03-18T06:23:34.080892",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2026-03-18T06:23:34.133458"
  },
  {
    "round": 4,
    "train_loss": 0.9923938412077214,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 0.9917313167056208,
      "mae": 0.9950340238561068,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    },
    "aggregation": {
      "num_clients": 1,
      "total_samples": 10
    },
    "client_metrics": [
      {
        "client_id": "android_client_520",
        "loss": 0.9923938412077214,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 13,
      "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_arm64",
        "manufacturer": "Google",
        "version": "16",
        "sdkInt": 36
      },
      "timestamp": "2026-03-18T06:23:34.786034",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2026-03-18T06:23:34.817792"
  },
  {
    "round": 5,
    "train_loss": 0.9923938412077214,
    "test_metrics": {
      "accuracy": 0.0,
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"Recall@10": 0.0,
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      "sdkInt": 36
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    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
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    "Precision@10": 0.0,
    "Recall@10": 0.0,
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      "epochs": 1
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    "Precision@10": 0.0,
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    "Precision@10": 0.0,
    "Recall@10": 0.0,
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    "mae": 0.9593951932586049,
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    "NDCG@10": 0.0,
    "Precision@10": 0.0,
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    "total_samples": 15
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      "samples": 15,
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    "Precision@10": 0.0,
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    "total_samples": 15
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    "Precision@10": 0.0,
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    "total_samples": 15
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      "epochs": 1
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    "NDCG@10": 0.0,
    "Precision@10": 0.0,
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    "samples": 15
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    "total_samples": 15
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      "samples": 15,
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  ],
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        "mae": 0.9593951932586049,
        "Hit@10": 0.0,
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        "Precision@10": 0.0,
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        "total_samples": 15
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        "NDCG@10": 0.0,
        "Precision@10": 0.0,
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        "total_samples": 15
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        "mae": 0.9593951932586051,
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        "NDCG@10": 0.0,
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    "manufacturer": "Google",
    "version": "16",
    "sdkInt": 36
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{
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    "NDCG@10": 0.0,
    "Precision@10": 0.0,
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    "total_samples": 15
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      "samples": 15,
      "epochs": 1
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      "manufacturer": "Google",
      "version": "16",
      "sdkInt": 36
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{
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    "NDCG@10": 0.0,
    "Precision@10": 0.0,
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    "battery_drain": 0.0,
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      "model": "sdk_gphone64_arm64",
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        "sdkInt": 36
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{
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        "Precision@10": 0.0,
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            "manufacturer": "Google",
            "version": "16",
            "sdkInt": 36
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{
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  "Precision@10": 0.0,
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    "Precision@10": 0.0,
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    "Precision@10": 0.0,
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      "NDCG@10": 0.0,
      "Precision@10": 0.0,
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      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
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    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
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"Precision@10": 0.0,
"Recall@10": 0.0,
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  "total_samples": 18
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    "Precision@10": 0.0,
    "Recall@10": 0.0,
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}
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    "NDCG@10": 0.0,
    "Precision@10": 0.0,
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    "samples": 18
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    "total_samples": 18
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    "Precision@10": 0.0,
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    "total_samples": 18
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    "Precision@10": 0.0,
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    "Precision@10": 0.0,
    "Recall@10": 0.0,
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    "total_samples": 18
  },
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    },
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        "version": "16",
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      "Precision@10": 0.0,
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    "device_info": {
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      "model": "sdk_gphone64_arm64",
      "manufacturer": "Google",
      "version": "16",
      "sdkInt": 36
    },
    "timestamp": "2026-03-18T22:34:32.537825",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2026-03-18T22:34:32.618449"
},
"final_metrics": {
  "accuracy": 0.0,
  "mse": 0.9184057305222646,
  "mae": 0.956466859424305,
  "Hit@10": 0.0,
  "NDCG@10": 0.0,
  "Precision@10": 0.0,
  "Recall@10": 0.0,
  "final_train_loss": 0.9568706929976282,
  "avg_accuracy": 0.0,
  "avg_Hit@10": 0.0,
  "avg_NDCG@10": 0.0
},
"client_metrics": [
  {
    "client_id": "android_client_520",
    "total_rounds": 11,
    "total_samples": 10,
    "final_loss": 0.9923938412077216,
    "final_accuracy": 0.0,
    "timestamp": "2026-03-18T06:23:45.534544"
  },
  {
    "client_id": "android_client_520",
    "total_rounds": 22,
    "total_samples": 15,
    "final_loss": 0.9585907083365778,
    "final_accuracy": 0.0,
    "timestamp": "2026-03-18T06:25:07.932173"
  },
  {
    "client_id": "android_client_520",
    "total_rounds": 23,
    "total_samples": 16,
    "final_loss": 0.9568706929976282,
    "final_accuracy": 0.0,
    "timestamp": "2026-03-18T22:28:26.870775"
```

```
}
],
"mobile_metrics": [
  {
    "device_id": "google_sdk_gphone64_arm64",
    "total_rounds": 11,
    "total_battery_drain": 0.0,
    "total_training_time_ms": 0,
    "avg_battery_drain_per_round": 0.0,
    "avg_training_time_ms_per_round": 0.0,
    "timestamp": "2026-03-18T06:23:45.536023"
  },
  {
    "device_id": "google_sdk_gphone64_arm64",
    "total_rounds": 22,
    "total_battery_drain": 0.0,
    "total_training_time_ms": 0,
    "avg_battery_drain_per_round": 0.0,
    "avg_training_time_ms_per_round": 0.0,
    "timestamp": "2026-03-18T06:25:07.932259"
  },
  {
    "device_id": "google_sdk_gphone64_arm64",
    "total_rounds": 23,
    "total_battery_drain": 0.0,
    "total_training_time_ms": 0,
    "avg_battery_drain_per_round": 0.0,
    "avg_training_time_ms_per_round": 0.0,
    "timestamp": "2026-03-18T22:28:26.871286"
  }
],
"server_received_at": "2026-03-18T22:34:33.074729"
}
```

File: mobile_results/dp_inf_dim_16_clients_1_device_google_sdk_gphone64_arm64_t1773795201066_summary.csv

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients,Total_Samples,Training_Time_MS,Battery_Drain

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```

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```

File: mobile_results/dp_inf_dim_16_clients_1_device_google_sdk_gphone64_x86_64_t1764480689336.json

```

{
  "experiment_id": "dp_inf_dim_16_clients_1_device_google_sdk_gphone64_x86_64_t1764480689336",
  "timestamp": "2025-11-30T11:01:29.337186",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 16,
    "client_id": "android_client_7270",
    "device_id": "google_sdk_gphone64_x86_64",
    "server_url": "http://192.168.29.146:8000"
  },
  "rounds": [
    {
      "round": 0,
      "train_loss": 0.0,
      "test_metrics": {
        "accuracy": 0.0,
        "mse": 0.0,
        "mae": 0.0,
        "Hit@10": 0.0,
        "NDCG@10": 0.0,
        "Precision@10": 0.0,
        "Recall@10": 0.0
      },
      "aggregation": {
        "num_clients": 1,
        "total_samples": 0
      },
      "client_metrics": [
        {
          "client_id": "android_client_7270",
          "loss": 0.0,
          "samples": 0,
          "epochs": 0
        }
      ],
      "resource_metrics": {
        "battery_level": 100,
        "battery_drain": 0.0,
        "elapsed_seconds": 14,
        "device_info": {
          "platform": "Android",
          "model": "sdk_gphone64_x86_64",
          "manufacturer": "Google",
          "version": "15",
          "sdkInt": 35
        },
        "timestamp": "2025-11-30T11:01:43.620018",
        "cpu_percent": 15.0,
        "memory_mb": 50.0
      },
      "timestamp": "2025-11-30T11:01:43.780785"
    },
    {
      "round": 1,
      "train_loss": 0.9998308269768555,
      "test_metrics": {
        "accuracy": 0.0,
        "mse": 0.9998242840096119,
        "mae": 0.99991209000057422,
        "Hit@10": 0.0,
        "NDCG@10": 0.0,
        "Precision@10": 0.0,

```

```
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_7270",
      "loss": 0.9998308269768555,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 15,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_x86_64",
      "manufacturer": "Google",
      "version": "15",
      "sdkInt": 35
    },
    "timestamp": "2025-11-30T11:01:44.999789",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2025-11-30T11:01:45.168931"
},
{
  "round": 2,
  "train_loss": 0.9998308269768554,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998242840096119,
    "mae": 0.9999120900005742,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_7270",
      "loss": 0.9998308269768554,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 16,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_x86_64",
      "manufacturer": "Google",
      "version": "15",
      "sdkInt": 35
    },
    "timestamp": "2025-11-30T11:01:46.117730",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2025-11-30T11:01:46.290431"
},
```

```
{
  "round": 3,
  "train_loss": 0.9998308269768554,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998242840096119,
    "mae": 0.999912090005742,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_7270",
      "loss": 0.9998308269768554,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 17,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_x86_64",
      "manufacturer": "Google",
      "version": "15",
      "sdkInt": 35
    },
    "timestamp": "2025-11-30T11:01:47.190404",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2025-11-30T11:01:47.355588"
},
{
  "round": 4,
  "train_loss": 0.9998308269768555,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998242840096117,
    "mae": 0.999912090005742,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_7270",
      "loss": 0.9998308269768555,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 18,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_x86_64",
```

```
    "manufacturer": "Google",
    "version": "15",
    "sdkInt": 35
  },
  "timestamp": "2025-11-30T11:01:48.202088",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2025-11-30T11:01:48.342287"
},
{
  "round": 5,
  "train_loss": 0.9998308269768554,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998242840096119,
    "mae": 0.999912090005742,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_7270",
      "loss": 0.9998308269768554,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 19,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_x86_64",
      "manufacturer": "Google",
      "version": "15",
      "sdkInt": 35
    },
    "timestamp": "2025-11-30T11:01:49.167143",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2025-11-30T11:01:49.324384"
},
{
  "round": 6,
  "train_loss": 0.9998308269768554,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998242840096119,
    "mae": 0.9999120900057422,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_7270",
      "loss": 0.9998308269768554,
      "samples": 10,
```

```
    "epochs": 1
  },
  {
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 20,
      "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_x86_64",
        "manufacturer": "Google",
        "version": "15",
        "sdkInt": 35
      },
      "timestamp": "2025-11-30T11:01:50.137856",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2025-11-30T11:01:50.293700"
  },
  {
    "round": 7,
    "train_loss": 0.9998308269768555,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 0.9998242840096117,
      "mae": 0.999912090005742,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    },
    "aggregation": {
      "num_clients": 1,
      "total_samples": 10
    },
    "client_metrics": [
      {
        "client_id": "android_client_7270",
        "loss": 0.9998308269768555,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 21,
      "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_x86_64",
        "manufacturer": "Google",
        "version": "15",
        "sdkInt": 35
      },
      "timestamp": "2025-11-30T11:01:51.143013",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2025-11-30T11:01:51.285619"
  },
  {
    "round": 8,
    "train_loss": 0.9998308269768552,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 0.9998242840096119,
      "mae": 0.999912090005742,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    },
```

```
    },
    "aggregation": {
      "num_clients": 1,
      "total_samples": 10
    },
    "client_metrics": [
      {
        "client_id": "android_client_7270",
        "loss": 0.9998308269768552,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 22,
      "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_x86_64",
        "manufacturer": "Google",
        "version": "15",
        "sdkInt": 35
      },
      "timestamp": "2025-11-30T11:01:52.083615",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2025-11-30T11:01:52.233009"
  },
  {
    "round": 9,
    "train_loss": 0.9998308269768554,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 0.9998242840096119,
      "mae": 0.999912090005742,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    },
    "aggregation": {
      "num_clients": 1,
      "total_samples": 10
    },
    "client_metrics": [
      {
        "client_id": "android_client_7270",
        "loss": 0.9998308269768554,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 23,
      "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_x86_64",
        "manufacturer": "Google",
        "version": "15",
        "sdkInt": 35
      },
      "timestamp": "2025-11-30T11:01:53.074917",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2025-11-30T11:01:53.226124"
  },
  {
    "round": 0,
```



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"train_loss": 0.9998308269768554,
"test_metrics": {
  "accuracy": 0.0,
  "mse": 0.9998242840096119,
  "mae": 0.999912090005742,
  "Hit@10": 0.0,
  "NDCG@10": 0.0,
  "Precision@10": 0.0,
  "Recall@10": 0.0,
  "samples": 10
},
"aggregation": {
  "num_clients": 1,
  "total_samples": 10
},
"client_metrics": [
  {
    "client_id": "android_client_7270",
    "loss": 0.9998308269768554,
    "samples": 10,
    "epochs": 1
  }
],
"resource_metrics": {
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  "battery_drain": 0.0,
  "elapsed_seconds": 49,
  "device_info": {
    "platform": "Android",
    "model": "sdk_gphone64_x86_64",
    "manufacturer": "Google",
    "version": "15",
    "sdkInt": 35
  },
  "timestamp": "2025-11-30T11:02:18.943458",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2025-11-30T11:02:19.125376"
},
{
  "round": 1,
  "train_loss": 0.9998308269768555,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998242840096117,
    "mae": 0.9999120900057422,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_7270",
      "loss": 0.9998308269768555,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
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    "battery_drain": 0.0,
    "elapsed_seconds": 50,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_x86_64",
      "manufacturer": "Google",
      "version": "15",
```

```
    "sdkInt": 35
  },
  "timestamp": "2025-11-30T11:02:20.079165",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2025-11-30T11:02:20.228799"
},
{
  "round": 2,
  "train_loss": 0.9998308269768555,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998242840096117,
    "mae": 0.9999120900057422,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_7270",
      "loss": 0.9998308269768555,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 51,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_x86_64",
      "manufacturer": "Google",
      "version": "15",
      "sdkInt": 35
    }
  },
  "timestamp": "2025-11-30T11:02:21.070842",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2025-11-30T11:02:21.201995"
},
{
  "round": 3,
  "train_loss": 0.9998308269768552,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998242840096119,
    "mae": 0.9999120900057422,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_7270",
      "loss": 0.9998308269768552,
      "samples": 10,
      "epochs": 1
    }
  ]
}
```

```
    },
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 52,
      "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_x86_64",
        "manufacturer": "Google",
        "version": "15",
        "sdkInt": 35
      },
      "timestamp": "2025-11-30T11:02:22.068837",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2025-11-30T11:02:22.215471"
  },
  {
    "round": 4,
    "train_loss": 0.9998308269768554,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 0.9998242840096119,
      "mae": 0.9999120900057422,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    },
    "aggregation": {
      "num_clients": 1,
      "total_samples": 10
    },
    "client_metrics": [
      {
        "client_id": "android_client_7270",
        "loss": 0.9998308269768554,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 53,
      "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_x86_64",
        "manufacturer": "Google",
        "version": "15",
        "sdkInt": 35
      },
      "timestamp": "2025-11-30T11:02:23.046662",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2025-11-30T11:02:23.230707"
  },
  {
    "round": 5,
    "train_loss": 0.9998308269768554,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 0.9998242840096117,
      "mae": 0.9999120900057422,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    },
    "aggregation": {
```

```
"num_clients": 1,
"total_samples": 10
},
"client_metrics": [
  {
    "client_id": "android_client_7270",
    "loss": 0.9998308269768554,
    "samples": 10,
    "epochs": 1
  }
],
"resource_metrics": {
  "battery_level": 100,
  "battery_drain": 0.0,
  "elapsed_seconds": 54,
  "device_info": {
    "platform": "Android",
    "model": "sdk_gphone64_x86_64",
    "manufacturer": "Google",
    "version": "15",
    "sdkInt": 35
  },
  "timestamp": "2025-11-30T11:02:23.996764",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2025-11-30T11:02:24.135641"
},
{
  "round": 6,
  "train_loss": 0.9998308269768554,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998242840096117,
    "mae": 0.9999120900057422,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_7270",
      "loss": 0.9998308269768554,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
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    "battery_drain": 0.0,
    "elapsed_seconds": 55,
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      "model": "sdk_gphone64_x86_64",
      "manufacturer": "Google",
      "version": "15",
      "sdkInt": 35
    },
    "timestamp": "2025-11-30T11:02:24.921052",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2025-11-30T11:02:25.048099"
},
{
  "round": 7,
  "train_loss": 0.9998308269768554,
  "test_metrics": {
```

```
"accuracy": 0.0,
"mse": 0.9998242840096117,
"mae": 0.999912090005742,
"Hit@10": 0.0,
"NDCG@10": 0.0,
"Precision@10": 0.0,
"Recall@10": 0.0,
"samples": 10
},
"aggregation": {
  "num_clients": 1,
  "total_samples": 10
},
"client_metrics": [
  {
    "client_id": "android_client_7270",
    "loss": 0.9998308269768554,
    "samples": 10,
    "epochs": 1
  }
],
"resource_metrics": {
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  "battery_drain": 0.0,
  "elapsed_seconds": 56,
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    "model": "sdk_gphone64_x86_64",
    "manufacturer": "Google",
    "version": "15",
    "sdkInt": 35
  },
  "timestamp": "2025-11-30T11:02:25.881762",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2025-11-30T11:02:26.031096"
},
{
  "round": 8,
  "train_loss": 0.9998308269768554,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998242840096119,
    "mae": 0.999912090005742,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_7270",
      "loss": 0.9998308269768554,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 57,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_x86_64",
      "manufacturer": "Google",
      "version": "15",
      "sdkInt": 35
    }
  },
}
```

```

        "timestamp": "2025-11-30T11:02:26.839255",
        "cpu_percent": 15.0,
        "memory_mb": 50.0
    },
    "timestamp": "2025-11-30T11:02:26.972702"
},
{
    "round": 9,
    "train_loss": 0.9998308269768554,
    "test_metrics": {
        "accuracy": 0.0,
        "mse": 0.9998242840096119,
        "mae": 0.999912090005742,
        "Hit@10": 0.0,
        "NDCG@10": 0.0,
        "Precision@10": 0.0,
        "Recall@10": 0.0,
        "samples": 10
    },
    "aggregation": {
        "num_clients": 1,
        "total_samples": 10
    },
    "client_metrics": [
        {
            "client_id": "android_client_7270",
            "loss": 0.9998308269768554,
            "samples": 10,
            "epochs": 1
        }
    ],
    "resource_metrics": {
        "battery_level": 100,
        "battery_drain": 0.0,
        "elapsed_seconds": 58,
        "device_info": {
            "platform": "Android",
            "model": "sdk_gphone64_x86_64",
            "manufacturer": "Google",
            "version": "15",
            "sdkInt": 35
        },
        "timestamp": "2025-11-30T11:02:27.811783",
        "cpu_percent": 15.0,
        "memory_mb": 50.0
    },
    "timestamp": "2025-11-30T11:02:27.938129"
}
],
"final_metrics": {},
"client_metrics": [],
"mobile_metrics": [],
"server_received_at": "2025-11-30T11:02:32.759551"
}

```

File: mobile_results/dp_inf_dim_16_clients_1_device_google_sdk_gphone64_x86_64_t1764480689336_summary.csv

Round, Train_Loss, NDCG@10, Hit@10, Precision@10, Recall@10, MSE, MAE, Accuracy, Num_Clients, Total_Samples, Training_Time_MS, Battery_Drain

```

0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,1,0,,0.0
1,0.9998308269768555,0.0,0.0,0.0,0.0,0.9998242840096119,0.9999120900057422,0.0,1,10,,0.0
2,0.9998308269768554,0.0,0.0,0.0,0.0,0.9998242840096119,0.999912090005742,0.0,1,10,,0.0
3,0.9998308269768554,0.0,0.0,0.0,0.0,0.9998242840096119,0.999912090005742,0.0,1,10,,0.0
4,0.9998308269768555,0.0,0.0,0.0,0.0,0.9998242840096117,0.999912090005742,0.0,1,10,,0.0
5,0.9998308269768554,0.0,0.0,0.0,0.0,0.9998242840096119,0.999912090005742,0.0,1,10,,0.0
6,0.9998308269768554,0.0,0.0,0.0,0.0,0.9998242840096119,0.9999120900057422,0.0,1,10,,0.0
7,0.9998308269768555,0.0,0.0,0.0,0.0,0.9998242840096117,0.999912090005742,0.0,1,10,,0.0
8,0.9998308269768552,0.0,0.0,0.0,0.0,0.9998242840096119,0.999912090005742,0.0,1,10,,0.0
9,0.9998308269768554,0.0,0.0,0.0,0.0,0.9998242840096119,0.999912090005742,0.0,1,10,,0.0
0,0.9998308269768554,0.0,0.0,0.0,0.0,0.9998242840096119,0.999912090005742,0.0,1,10,,0.0
1,0.9998308269768555,0.0,0.0,0.0,0.0,0.9998242840096117,0.9999120900057422,0.0,1,10,,0.0
2,0.9998308269768555,0.0,0.0,0.0,0.0,0.9998242840096117,0.9999120900057422,0.0,1,10,,0.0
3,0.9998308269768552,0.0,0.0,0.0,0.0,0.9998242840096119,0.9999120900057422,0.0,1,10,,0.0
4,0.9998308269768554,0.0,0.0,0.0,0.0,0.9998242840096119,0.9999120900057422,0.0,1,10,,0.0

```

```

5,0.9998308269768554,0.0,0.0,0.0,0.0,0.9998242840096117,0.9999120900057422,0.0,1,10,,0.0
6,0.9998308269768554,0.0,0.0,0.0,0.0,0.9998242840096117,0.9999120900057422,0.0,1,10,,0.0
7,0.9998308269768554,0.0,0.0,0.0,0.0,0.9998242840096117,0.9999120900057422,0.0,1,10,,0.0
8,0.9998308269768554,0.0,0.0,0.0,0.0,0.9998242840096119,0.9999120900057422,0.0,1,10,,0.0
9,0.9998308269768554,0.0,0.0,0.0,0.0,0.9998242840096119,0.9999120900057422,0.0,1,10,,0.0
FINAL,,,,,,,,,

```

File: mobile_results/dp_inf_dim_16_clients_1_device_google_sdk_gphone64_x86_64_t1764495775358.json

```

{
  "experiment_id": "dp_inf_dim_16_clients_1_device_google_sdk_gphone64_x86_64_t1764495775358",
  "timestamp": "2025-11-30T15:12:55.359890",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 16,
    "client_id": "android_client_3511",
    "device_id": "google_sdk_gphone64_x86_64",
    "server_url": "http://192.168.29.146:8000"
  },
  "rounds": [
    {
      "round": 0,
      "train_loss": 0.0,
      "test_metrics": {
        "accuracy": 0.0,
        "mse": 0.0,
        "mae": 0.0,
        "Hit@10": 0.0,
        "NDCG@10": 0.0,
        "Precision@10": 0.0,
        "Recall@10": 0.0
      },
      "aggregation": {
        "num_clients": 1,
        "total_samples": 0
      },
      "client_metrics": [
        {
          "client_id": "android_client_3511",
          "loss": 0.0,
          "samples": 0,
          "epochs": 0
        }
      ],
      "resource_metrics": {
        "battery_level": 100,
        "battery_drain": 0.0,
        "elapsed_seconds": 6,
        "device_info": {
          "platform": "Android",
          "model": "sdk_gphone64_x86_64",
          "manufacturer": "Google",
          "version": "16",
          "sdkInt": 36
        },
        "timestamp": "2025-11-30T15:13:02.060511",
        "cpu_percent": 15.0,
        "memory_mb": 50.0
      },
      "timestamp": "2025-11-30T15:13:02.218085"
    },
    {
      "round": 1,
      "train_loss": 0.999905153416518,
      "test_metrics": {
        "accuracy": 0.0,
        "mse": 0.9998990150927533,
        "mae": 0.999949461917601,
        "Hit@10": 0.0,
        "NDCG@10": 0.0,
        "Precision@10": 0.0,
        "Recall@10": 0.0,
        "samples": 10
      }
    }
  ]
}

```

```
    },
    "aggregation": {
      "num_clients": 1,
      "total_samples": 10
    },
    "client_metrics": [
      {
        "client_id": "android_client_3511",
        "loss": 0.999905153416518,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 8,
      "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_x86_64",
        "manufacturer": "Google",
        "version": "16",
        "sdkInt": 36
      },
      "timestamp": "2025-11-30T15:13:03.853133",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2025-11-30T15:13:04.068787"
  },
  {
    "round": 2,
    "train_loss": 0.9999051534165181,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 0.9998990150927531,
      "mae": 0.999949461917601,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    },
    "aggregation": {
      "num_clients": 1,
      "total_samples": 10
    },
    "client_metrics": [
      {
        "client_id": "android_client_3511",
        "loss": 0.9999051534165181,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 10,
      "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_x86_64",
        "manufacturer": "Google",
        "version": "16",
        "sdkInt": 36
      },
      "timestamp": "2025-11-30T15:13:05.602396",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2025-11-30T15:13:05.794214"
  },
  {
    "round": 3,
```



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"train_loss": 0.999905153416518,
"test_metrics": {
  "accuracy": 0.0,
  "mse": 0.9998990150927533,
  "mae": 0.999949461917601,
  "Hit@10": 0.0,
  "NDCG@10": 0.0,
  "Precision@10": 0.0,
  "Recall@10": 0.0,
  "samples": 10
},
"aggregation": {
  "num_clients": 1,
  "total_samples": 10
},
"client_metrics": [
  {
    "client_id": "android_client_3511",
    "loss": 0.999905153416518,
    "samples": 10,
    "epochs": 1
  }
],
"resource_metrics": {
  "battery_level": 100,
  "battery_drain": 0.0,
  "elapsed_seconds": 11,
  "device_info": {
    "platform": "Android",
    "model": "sdk_gphone64_x86_64",
    "manufacturer": "Google",
    "version": "16",
    "sdkInt": 36
  },
  "timestamp": "2025-11-30T15:13:06.688705",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2025-11-30T15:13:06.820583"
},
{
  "round": 4,
  "train_loss": 0.9999051534165178,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998990150927531,
    "mae": 0.999949461917601,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_3511",
      "loss": 0.9999051534165178,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 12,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_x86_64",
      "manufacturer": "Google",
      "version": "16",
```

```
    "sdkInt": 36
  },
  "timestamp": "2025-11-30T15:13:07.702132",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2025-11-30T15:13:07.818781"
},
{
  "round": 5,
  "train_loss": 0.9999051534165181,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998990150927533,
    "mae": 0.999949461917601,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_3511",
      "loss": 0.9999051534165181,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 13,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_x86_64",
      "manufacturer": "Google",
      "version": "16",
      "sdkInt": 36
    }
  },
  "timestamp": "2025-11-30T15:13:08.724818",
  "cpu_percent": 15.0,
  "memory_mb": 50.0
},
"timestamp": "2025-11-30T15:13:08.881306"
},
{
  "round": 6,
  "train_loss": 0.999905153416518,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998990150927531,
    "mae": 0.999949461917601,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_3511",
      "loss": 0.999905153416518,
      "samples": 10,
      "epochs": 1
    }
  ]
}
```

```
    },
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 14,
      "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_x86_64",
        "manufacturer": "Google",
        "version": "16",
        "sdkInt": 36
      },
      "timestamp": "2025-11-30T15:13:09.749715",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2025-11-30T15:13:09.885667"
  },
  {
    "round": 7,
    "train_loss": 0.999905153416518,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 0.9998990150927531,
      "mae": 0.999949461917601,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    },
    "aggregation": {
      "num_clients": 1,
      "total_samples": 10
    },
    "client_metrics": [
      {
        "client_id": "android_client_3511",
        "loss": 0.999905153416518,
        "samples": 10,
        "epochs": 1
      }
    ],
    "resource_metrics": {
      "battery_level": 100,
      "battery_drain": 0.0,
      "elapsed_seconds": 15,
      "device_info": {
        "platform": "Android",
        "model": "sdk_gphone64_x86_64",
        "manufacturer": "Google",
        "version": "16",
        "sdkInt": 36
      },
      "timestamp": "2025-11-30T15:13:10.719598",
      "cpu_percent": 15.0,
      "memory_mb": 50.0
    },
    "timestamp": "2025-11-30T15:13:10.836819"
  },
  {
    "round": 8,
    "train_loss": 0.999905153416518,
    "test_metrics": {
      "accuracy": 0.0,
      "mse": 0.9998990150927531,
      "mae": 0.999949461917601,
      "Hit@10": 0.0,
      "NDCG@10": 0.0,
      "Precision@10": 0.0,
      "Recall@10": 0.0,
      "samples": 10
    },
    "aggregation": {
```

```
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_3511",
      "loss": 0.999905153416518,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 16,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_x86_64",
      "manufacturer": "Google",
      "version": "16",
      "sdkInt": 36
    },
    "timestamp": "2025-11-30T15:13:11.664750",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2025-11-30T15:13:11.785589"
},
{
  "round": 9,
  "train_loss": 0.9999051534165181,
  "test_metrics": {
    "accuracy": 0.0,
    "mse": 0.9998990150927531,
    "mae": 0.9999494619176008,
    "Hit@10": 0.0,
    "NDCG@10": 0.0,
    "Precision@10": 0.0,
    "Recall@10": 0.0,
    "samples": 10
  },
  "aggregation": {
    "num_clients": 1,
    "total_samples": 10
  },
  "client_metrics": [
    {
      "client_id": "android_client_3511",
      "loss": 0.9999051534165181,
      "samples": 10,
      "epochs": 1
    }
  ],
  "resource_metrics": {
    "battery_level": 100,
    "battery_drain": 0.0,
    "elapsed_seconds": 17,
    "device_info": {
      "platform": "Android",
      "model": "sdk_gphone64_x86_64",
      "manufacturer": "Google",
      "version": "16",
      "sdkInt": 36
    },
    "timestamp": "2025-11-30T15:13:12.584036",
    "cpu_percent": 15.0,
    "memory_mb": 50.0
  },
  "timestamp": "2025-11-30T15:13:12.678029"
}
],
"final_metrics": {},
"client_metrics": [],
"mobile_metrics": [],
```

```
"server_received_at": "2025-11-30T15:13:16.376746"
}
```

File: mobile_results/dp_inf_dim_16_clients_1_device_google_sdk_gphone64_x86_64_t1764495775358_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients>Total_Samples,Training_
Time_MS,Battery_Drain
0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,1,0,,0.0
1,0.999905153416518,0.0,0.0,0.0,0.0,0.9998990150927533,0.999949461917601,0.0,1,10,,0.0
2,0.9999051534165181,0.0,0.0,0.0,0.0,0.9998990150927531,0.999949461917601,0.0,1,10,,0.0
3,0.999905153416518,0.0,0.0,0.0,0.0,0.9998990150927533,0.999949461917601,0.0,1,10,,0.0
4,0.9999051534165178,0.0,0.0,0.0,0.0,0.9998990150927531,0.999949461917601,0.0,1,10,,0.0
5,0.9999051534165181,0.0,0.0,0.0,0.0,0.9998990150927533,0.999949461917601,0.0,1,10,,0.0
6,0.999905153416518,0.0,0.0,0.0,0.0,0.9998990150927531,0.999949461917601,0.0,1,10,,0.0
7,0.999905153416518,0.0,0.0,0.0,0.0,0.9998990150927531,0.999949461917601,0.0,1,10,,0.0
8,0.999905153416518,0.0,0.0,0.0,0.0,0.9998990150927531,0.999949461917601,0.0,1,10,,0.0
9,0.9999051534165181,0.0,0.0,0.0,0.0,0.9998990150927531,0.9999494619176008,0.0,1,10,,0.0
FINAL,,,,,,,,,
```

File: python_scripts/convert_all_dart_files_to_pdf.py

```
import argparse
import os
from pathlib import Path

try:
    from reportlab.lib.pagesizes import A4
    from reportlab.pdfbase.pdfmetrics import stringWidth
    from reportlab.pdfgen import canvas
except ImportError as exc:
    raise SystemExit(
        "Missing dependency: reportlab. Install it with `pip install reportlab` and try again."
    ) from exc

SUPPORTED_EXTENSIONS = {
    ".dart", ".py", ".json", ".csv", ".md"
}

SKIP_DIRECTORIES = {
    ".git", "__pycache__", "node_modules", ".idea", ".vscode",
    ".venv", "venv", "build", "dist"
}

PAGE_WIDTH, PAGE_HEIGHT = A4
LEFT_MARGIN = 40
RIGHT_MARGIN = 40
TOP_MARGIN = 50
BOTTOM_MARGIN = 40
TITLE_FONT = "Helvetica-Bold"
BODY_FONT = "Courier"
TITLE_FONT_SIZE = 14
HEADING_FONT_SIZE = 10
BODY_FONT_SIZE = 8
LINE_HEIGHT = 10

def parse_arguments():
    parser = argparse.ArgumentParser(
        description="Export all supported code/text files in a folder to a single PDF."
    )
    parser.add_argument(
        "source_directory",
        nargs="?",
        help="Absolute or relative path to the folder containing code files.",
    )
    parser.add_argument(
        "output_directory",
        nargs="?",
        help="Directory where the generated PDF should be saved.",
    )
    parser.add_argument(
        "--output-name",
        default=None,
        help="Name of the generated PDF file without extension.",
    )
```

```
)
return parser.parse_args()

def prompt_for_missing_values(args):
    source_directory = args.source_directory or input(
        "Enter the absolute path to the folder containing the code files: "
    ).strip()
    output_directory = args.output_directory or input(
        "Enter the absolute path to the folder where you want to save the PDF: "
    ).strip()
    output_name = args.output_name or input(
        "Enter the name of the PDF file (without extension): "
    ).strip()
    return source_directory, output_directory, output_name

def normalize_output_name(output_name):
    cleaned_name = output_name.strip() or "combined_code_export"
    return cleaned_name if cleaned_name.lower().endswith(".pdf") else f"{cleaned_name}.pdf"

def iter_supported_files(source_directory):
    for root, dirs, files in os.walk(source_directory):
        dirs[:] = sorted(directory for directory in dirs if directory not in SKIP_DIRECTORIES)

        for file_name in sorted(files):
            file_path = Path(root) / file_name
            if file_path.suffix.lower() in SUPPORTED_EXTENSIONS:
                yield file_path, file_path.relative_to(source_directory)

def read_file_content(file_path):
    with open(file_path, "r", encoding="utf-8", errors="replace") as file_handle:
        return file_handle.read()

def wrap_text_line(text, font_name, font_size, max_width):
    expanded_text = text.expandtabs(4)
    if not expanded_text:
        return [""]

    wrapped_lines = []
    current_line = ""

    for character in expanded_text:
        candidate = f"{current_line}{character}"
        if current_line and stringWidth(candidate, font_name, font_size) > max_width:
            wrapped_lines.append(current_line)
            current_line = character
        else:
            current_line = candidate

    if current_line:
        wrapped_lines.append(current_line)

    return wrapped_lines

def start_page(pdf_canvas, page_number, pdf_title):
    pdf_canvas.setFont(TITLE_FONT, TITLE_FONT_SIZE)
    pdf_canvas.drawString(LEFT_MARGIN, PAGE_HEIGHT - TOP_MARGIN + 12, pdf_title)
    pdf_canvas.setFont("Helvetica", 9)
    pdf_canvas.drawRightString(PAGE_WIDTH - RIGHT_MARGIN, PAGE_HEIGHT - TOP_MARGIN + 12, f"Page {page_number}")
    pdf_canvas.line(LEFT_MARGIN, PAGE_HEIGHT - TOP_MARGIN + 6, PAGE_WIDTH - RIGHT_MARGIN, PAGE_HEIGHT - TOP_MARGIN + 6)
    return PAGE_HEIGHT - TOP_MARGIN - 8

MAX_PDF_SIZE_BYTES = 1 * 1024 * 1024 # 1 MB
MAX_PAGES_PER_PDF = 40 # ~5MB depending on content (adjust if needed)
def create_pdf_from_files(file_entries, pdf_file_path, source_directory):
    pdf_title = f"Code Export: {Path(source_directory).name}"
```

```

base_name = pdf_file_path.stem
output_dir = pdf_file_path.parent

TOTAL_PARTS = 19
total_files = len(file_entries)

if total_files == 0:
    return

# Compute chunk size (balanced split)
chunk_size = (total_files + TOTAL_PARTS - 1) // TOTAL_PARTS # ceil division

usable_width = PAGE_WIDTH - LEFT_MARGIN - RIGHT_MARGIN

for part_number in range(TOTAL_PARTS):
    start_idx = part_number * chunk_size
    end_idx = min(start_idx + chunk_size, total_files)

    if start_idx >= total_files:
        break # no more files

    part_entries = file_entries[start_idx:end_idx]

    part_path = output_dir / f"{base_name}_part{part_number + 1}.pdf"
    pdf_canvas = canvas.Canvas(str(part_path), pagesize=A4)
    pdf_canvas.setTitle(f"{pdf_title} (Part {part_number + 1})")

    page_number = 1
    y_position = start_page(pdf_canvas, page_number, pdf_title)

    # Header
    pdf_canvas.setFont("Helvetica", 9)
    pdf_canvas.drawString(LEFT_MARGIN, y_position, f"Source folder: {Path(source_directory).resolve()}")
)

    y_position -= LINE_HEIGHT
    pdf_canvas.drawString(LEFT_MARGIN, y_position, f"Files in this part: {len(part_entries)}")
    y_position -= LINE_HEIGHT * 2

    for file_path, relative_path in part_entries:

        if y_position <= BOTTOM_MARGIN + (LINE_HEIGHT * 4):
            pdf_canvas.showPage()
            page_number += 1
            y_position = start_page(pdf_canvas, page_number, pdf_title)

        pdf_canvas.setFont(TITLE_FONT, HEADING_FONT_SIZE)
        pdf_canvas.drawString(LEFT_MARGIN, y_position, f"File: {relative_path}")
        y_position -= LINE_HEIGHT

        pdf_canvas.setFont("Helvetica", 8)
        pdf_canvas.drawString(LEFT_MARGIN, y_position, "-" * 110)
        y_position -= LINE_HEIGHT

        pdf_canvas.setFont(BODY_FONT, BODY_FONT_SIZE)

        file_content = read_file_content(file_path)
        content_lines = file_content.splitlines() or [""]

        for line in content_lines:
            for wrapped_line in wrap_text_line(line, BODY_FONT, BODY_FONT_SIZE, usable_width):

                if y_position <= BOTTOM_MARGIN + LINE_HEIGHT:
                    pdf_canvas.showPage()
                    page_number += 1
                    y_position = start_page(pdf_canvas, page_number, pdf_title)
                    pdf_canvas.setFont(BODY_FONT, BODY_FONT_SIZE)

                pdf_canvas.drawString(LEFT_MARGIN, y_position, wrapped_line)
                y_position -= LINE_HEIGHT

            y_position -= LINE_HEIGHT

pdf_canvas.save()

```

```

    print(f"Created: {part_path}")

def main(source_directory, output_directory, output_name):
    source_directory = Path(source_directory).expanduser().resolve()
    output_directory = Path(output_directory).expanduser().resolve()

    if not source_directory.is_dir():
        print(f"Error: '{source_directory}' is not a valid directory.")
        return 1

    output_directory.mkdir(parents=True, exist_ok=True)

    file_entries = list(iter_supported_files(source_directory))
    if not file_entries:
        print(f"No supported code or text files found in '{source_directory}'.")
        return 1

    pdf_file_path = output_directory / normalize_output_name(output_name)
    create_pdf_from_files(file_entries, pdf_file_path, source_directory)
    print(f"Successfully created PDF: '{pdf_file_path}'")
    return 0

if __name__ == "__main__":
    cli_args = parse_arguments()
    source_dir, output_dir, output_name = prompt_for_missing_values(cli_args)
    raise SystemExit(main(source_dir, output_dir, output_name))

```

File: results/attack_evaluation_summary.json

```

{
  "inf": {
    "mia": {
      "train_accuracy": 1.0,
      "auc": 0.54805,
      "accuracy": 0.545,
      "true_positive_rate": 0.48,
      "false_positive_rate": 0.39
    },
    "inversion": {
      "top_k_accuracy": 0.0,
      "top_k": 10,
      "num_users_evaluated": 50
    }
  },
  "8": {
    "mia": {
      "train_accuracy": 1.0,
      "auc": 0.53355,
      "accuracy": 0.515,
      "true_positive_rate": 0.6,
      "false_positive_rate": 0.57
    },
    "inversion": {
      "top_k_accuracy": 0.0,
      "top_k": 10,
      "num_users_evaluated": 50
    }
  },
  "4": {
    "mia": {
      "train_accuracy": 1.0,
      "auc": 0.5024500000000001,
      "accuracy": 0.505,
      "true_positive_rate": 0.61,
      "false_positive_rate": 0.6
    },
    "inversion": {
      "top_k_accuracy": 0.0,
      "top_k": 10,
      "num_users_evaluated": 50
    }
  },
}

```



```
"2": {
  "mia": {
    "train_accuracy": 1.0,
    "auc": 0.5004,
    "accuracy": 0.5,
    "true_positive_rate": 0.64,
    "false_positive_rate": 0.64
  },
  "inversion": {
    "top_k_accuracy": 0.0,
    "top_k": 10,
    "num_users_evaluated": 50
  }
},
"1": {
  "mia": {
    "train_accuracy": 1.0,
    "auc": 0.48584999999999995,
    "accuracy": 0.51,
    "true_positive_rate": 0.62,
    "false_positive_rate": 0.6
  },
  "inversion": {
    "top_k_accuracy": 0.0,
    "top_k": 10,
    "num_users_evaluated": 50
  }
}
}
```

File: results/centralized_baseline.json

```
{
  "experiment_id": "centralized_baseline",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_epochs": 50,
    "learning_rate": 0.01,
    "batch_size": 64,
    "seed": 42,
    "train_samples": 80000,
    "test_samples": 20000
  },
  "train_losses": [
    13.744499945831299,
    13.733649127197266,
    13.7224368309021,
    13.710547796630859,
    13.697592328643799,
    13.683081722259521,
    13.666377948760987,
    13.646623229217528,
    13.622682015228271,
    13.593017753601075,
    13.555565504455567,
    13.507558028411864,
    13.445319319915772,
    13.364007292175293,
    13.257323151397705,
    13.117280011749267,
    12.93408091430664,
    12.696154238128662,
    12.390833408355713,
    12.005739274597168,
    11.53115025177002,
    10.963267489624023,
    10.307968677902222,
    9.583248323822021,
    8.81859575920105,
    8.050384371566773,
    7.314124571609497,
    6.636849789810181,

```

```
6.033216427993774,
5.506068143081665,
5.050083262443542,
4.655981608772278,
4.3136439014434815,
4.0142961641311645,
3.7506541988372804,
3.5169398542404173,
3.3086113360404967,
3.1220811665534973,
2.954416963863373,
2.803191617488861,
2.6663891534805297,
2.5422803891181944,
2.4293844024658204,
2.3264532668113707,
2.2323575150489807,
2.146175816440582,
2.0670647891044616,
1.9942732004165649,
1.9271537218093873,
1.865168280506134
],
"test_metrics_history": [
  {
    "epoch": 10,
    "metrics": {
      "mse": 13.622893885803222,
      "mae": 3.5133981464385986,
      "accuracy": 0.0,
      "samples": 20000,
      "NDCG@10": 0.06899745675111287,
      "Hit@10": 0.1,
      "Precision@10": 0.011000000000000001,
      "Recall@10": 0.00757346677004221,
      "num_users_evaluated": 100
    }
  },
  {
    "epoch": 20,
    "metrics": {
      "mse": 11.968002226257324,
      "mae": 3.2785331844329835,
      "accuracy": 0.0029,
      "samples": 20000,
      "NDCG@10": 0.24605057094704916,
      "Hit@10": 0.43,
      "Precision@10": 0.057999999999999996,
      "Recall@10": 0.027724766988414317,
      "num_users_evaluated": 100
    }
  },
  {
    "epoch": 30,
    "metrics": {
      "mse": 5.515742160415649,
      "mae": 2.0133992692947387,
      "accuracy": 0.0357,
      "samples": 20000,
      "NDCG@10": 0.28360105825565596,
      "Hit@10": 0.46,
      "Precision@10": 0.071000000000000001,
      "Recall@10": 0.035696776316918345,
      "num_users_evaluated": 100
    }
  },
  {
    "epoch": 40,
    "metrics": {
      "mse": 2.956100328826904,
      "mae": 1.3657585359573363,
      "accuracy": 0.04945,
      "samples": 20000,
      "NDCG@10": 0.2798283914482264,
```

```
    "Hit@10": 0.49,  
    "Precision@10": 0.072000000000000001,  
    "Recall@10": 0.04163180844534197,  
    "num_users_evaluated": 100  
  },  
  {  
    "epoch": 50,  
    "metrics": {  
      "mse": 2.0449277555465697,  
      "mae": 1.1063344123840333,  
      "accuracy": 0.05495,  
      "samples": 20000,  
      "NDCG@10": 0.22501065427109246,  
      "Hit@10": 0.38,  
      "Precision@10": 0.056000000000000001,  
      "Recall@10": 0.03393271862353286,  
      "num_users_evaluated": 100  
    }  
  },  
  {  
    "final_metrics": {  
      "mse": 2.0449277555465697,  
      "mae": 1.1063344123840333,  
      "accuracy": 0.05495,  
      "samples": 20000,  
      "NDCG@10": 0.22501065427109246,  
      "Hit@10": 0.38,  
      "Precision@10": 0.056000000000000001,  
      "Recall@10": 0.03393271862353286,  
      "num_users_evaluated": 100  
    }  
  }  
}
```